

Project Name:

CMOS 180nm Analog Comparator for Ambient Light Detection in Wearable Assistive Devices for Visually Impaired Patients

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1. Project Motivation & Clinical Scope

Rise of Visual Impairment Globally: Visual impairment affects over **2.2 billion people** worldwide, with at least 1 billion cases preventable or unaddressed [1]. Among these, approximately 36 million individuals are completely blind, facing life-threatening hazards in ambient light transitions.

Shift to Wearable Assistive Electronics: The healthcare paradigm is moving from hospital-based intervention to continuous, non-invasive, wearable monitoring. For visually impaired patients, real-time ambient light awareness directly reduces falls, supports circadian rhythm regulation, and enables safer navigation [2], [3].

The Wearable Challenge: Commercial assistive devices (LiDAR glasses, smartphone-based apps, ultrasonic aids) suffer from milliwatt-to-watt power budgets, bulky form factors, and fixed detection thresholds that fail across diverse lighting environments [4], [5].

Clinical Scope: This project proposes a **fully integrated CMOS analog comparator** operating at 1.8 V that provides real-time binary ambient light classification (high visibility / low visibility) for wearable skin-patch deployment. The circuit achieves sub-50 μW power consumption, adaptive threshold generation via a resistive divider, and a clean digital output for direct actuation of haptic or audio feedback — enabling reliable, always-on light awareness in ultra-low-power, small-form-factor wearable devices.

2. Light Signal Characterization & Design Requirements

2.1 Amplitude & Dynamic Range

Photodiode Output Range: 10 mV (dim indoor, ~ 10 lux) to 1.6 V (bright outdoor, $\sim 100,000$ lux) [6]

Typical Indoor Value: 0.1–0.4 V (office/hospital ward, 100–500 lux)

Threshold Voltage (V_{thresh}): Adaptively set to $V_{\text{max}} / 2$ via resistive divider; tracks ambient maximum in real time

The differential pair comparator must resolve input differences as small as **10 mV** without clipping across the full 0 V–VDD input range

2.2 Frequency Range (Bandwidth)

Ambient light variation rate: DC to ~ 10 Hz (walking pace, indoor-outdoor transitions) [7]

Minimum comparator bandwidth: > 1 kHz (provides $100\times$ margin over signal bandwidth)

Propagation delay requirement: < 1 ms (imperceptible latency for haptic feedback)

Defines the speed requirement for the differential pair and output inverter stages

2.3 Unwanted Components & Noise Sources

Photodiode dark current: $\sim 1\text{--}10$ nA at room temperature — converted to < 1 mV offset by TIA [8]

Transistor mismatch (M1/M2 offset): < 5 mV with common-centroid layout [9]

Supply ripple (VDD noise): < 10 mV with on-chip bypass capacitor

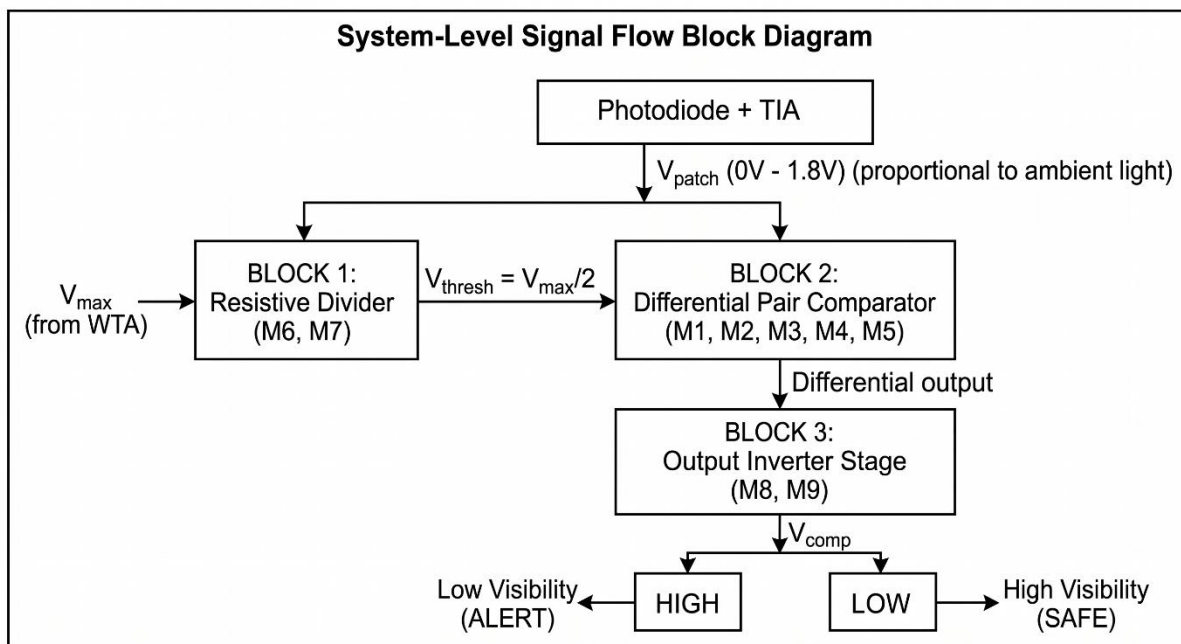
Flicker (1/f) noise: Dominant below 100 Hz; mitigated by operating the comparator in strong inversion (large W/L for M1, M2)

Requires careful layout matching of the differential input pair (M1, M2) and active load (M3, M4) to achieve reliable switching near the threshold

3. Proposed Comparator Architecture: System Top-Level

The proposed system is a **three-block CMOS analog comparator chain** that processes a photodiode-derived voltage and produces a binary digital output indicating ambient light level. The architecture is intentionally minimal — 9 transistors total — to achieve the lowest possible power consumption while maintaining robust switching performance.

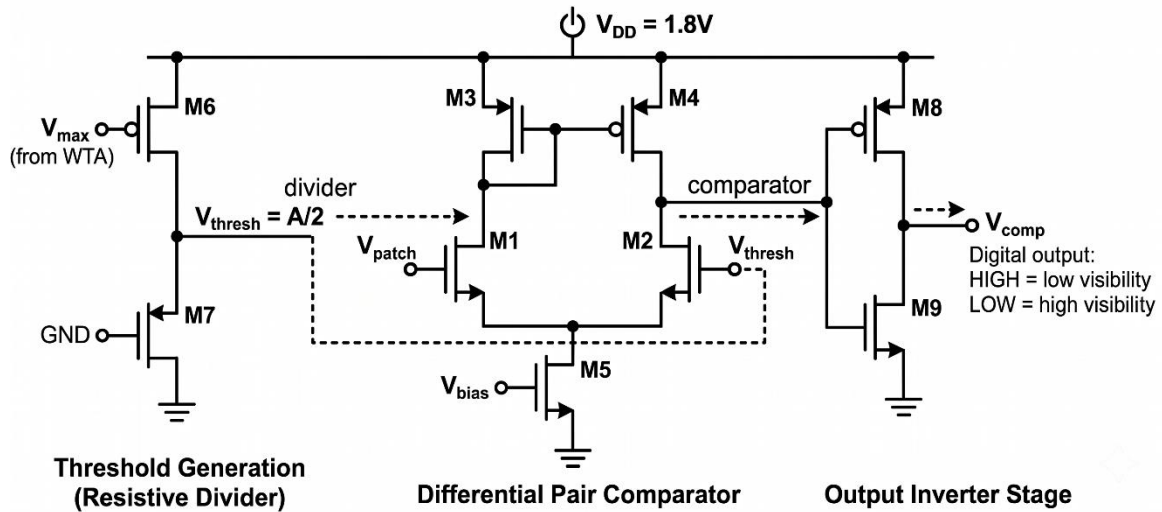
System-Level Signal Flow



Block 1 — Resistive Divider (Threshold Generator): Generates adaptive reference voltage $V_{thresh} = V_{max} / 2$ using a matched PMOS–NMOS voltage divider (M6, M7). V_{max} is supplied by a Winner-Take-All (WTA) preprocessing block that tracks the peak photodiode voltage.

Block 2 — Differential Pair Comparator: Compares V_{patch} (photodiode signal) against V_{thresh} (adaptive reference). Uses a symmetric NMOS differential pair (M1, M2) with PMOS current-mirror active load (M3, M4) and NMOS tail current source (M5). Provides > 20 dB open-loop voltage gain.

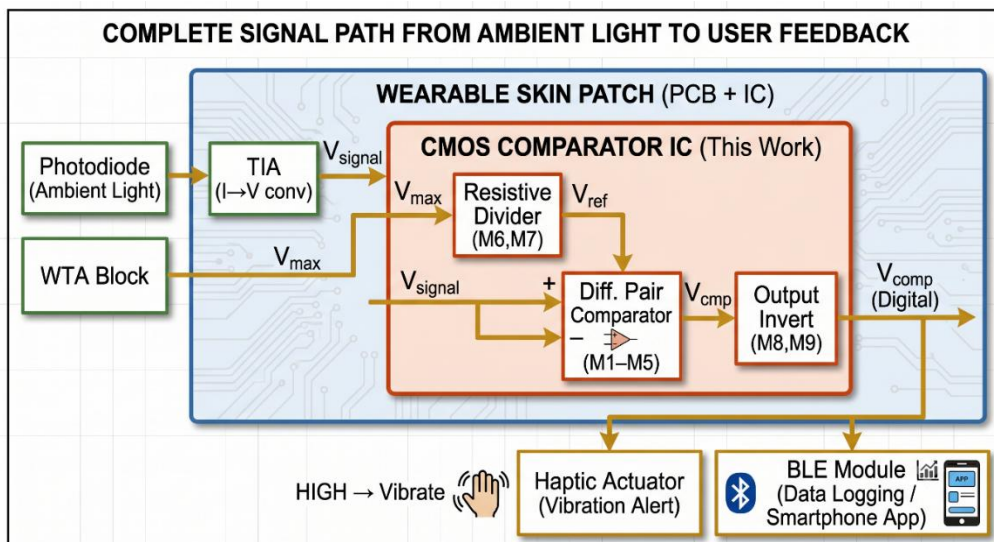
Block 3 — Output Inverter Stage: Converts the analog differential output to a clean rail-to-rail digital signal (0 V / 1.8 V) using a minimum-length CMOS inverter (M8, M9). The output directly drives a haptic actuator, audio transducer, or BLE module.



4. Application Diagrams

4.1 Wearable Light Awareness System — 3-Block Comparator Chain

The figure below illustrates the complete signal path from ambient light to user feedback through the proposed three-block CMOS comparator.



4.2 Clinical Use-Case Scenario

Ambient Light and System Response Scenarios

Scenario	Ambient Light	V_patch	V_thresh	V_comp	System Response
Bright outdoor (100k lux)	High	1.4 V	0.8 V	LOW	No alert — safe

Office indoor (500 lux)	Moderate	0.5 V	0.8 V	HIGH	Gentle vibration
Dark corridor (< 10 lux)	Very low	0.05 V	0.2 V	HIGH	Continuous alert
Night (0 lux)	None	~0 V	~0 V	HIGH	Max alert

4.3 Block Diagram (As Designed)

The three-block architecture directly corresponds to the circuit implementation:



5. Literature Comparison

Ref.	Author & Year	Tech. Node	VDD (V)	Power	Application	Sensor Type	Response Time	Adaptive Threshold	Wearable	Key Limitation
[10]	Stancieri et al., 2022	180 nm CMOS	1.8 V	~500 μ W	Biomedical optoelectronics	Photodiode + TIA	~1 μ s	No (fixed)	Partial	Fixed threshold; no wearable integration
[11]	Joo et al., 2023	Commercial MCU	3.3 V	~5 mW	Pulse oximetry (SpO2)	Reflective PPG	~10 ms	No	Yes (ring)	High power; no light classification
[12]	Shen et al., 2022	Raspberry Pi	5 V	~2.5 W	Blind pedestrian navigation	Camera + CNN	~33 ms	No	Partial (vest)	Very high power; bulky; no analog IC
[13]	Busaeed et al., 2022	FPGA + LiDAR	5 V	~3 W	Visually impaired mobility	LiDAR + DNN	~50 ms	No	Partial (glasses)	High power; expensive; no light sensing
[14]	Saha et al., 2021	Arduino MCU	5 V	~150 mW	Blind navigation aid	Ultrasonic + light	~100 ms	No	Partial (belt)	Discrete components; no CMOS IC
[15]	Zamora-Mejia et al., 2022	180 nm CMOS	1.5 V	~12 μ W	Biomedical UHF passive tag	RF energy harvesting	~1 ms	No	Yes (tag)	No light sensing; RF-only
[16]	Laababid et al., 2022	180 nm CMOS	1.8 V	~25 μ W	ECG wearable amplifier	Biopotential	~1 ms	No	Yes	ECG only; no comparator/light function
[17]	Erwinski et al., 2023	LoRa IoT node	3.3 V	~20 mW	Light pollution monitoring	Photodiode + MCU	~1 s	No	No (fixed node)	Not wearable; high power MCU

[18]	Sikka lagar et al., 2022	Microco ntroller	3.3 V	~100 mW	Wearable eye for blind	Camera + ultrasoni c	~200 ms	No	Yes (glasses)	High power; no CMOS IC; camera- based
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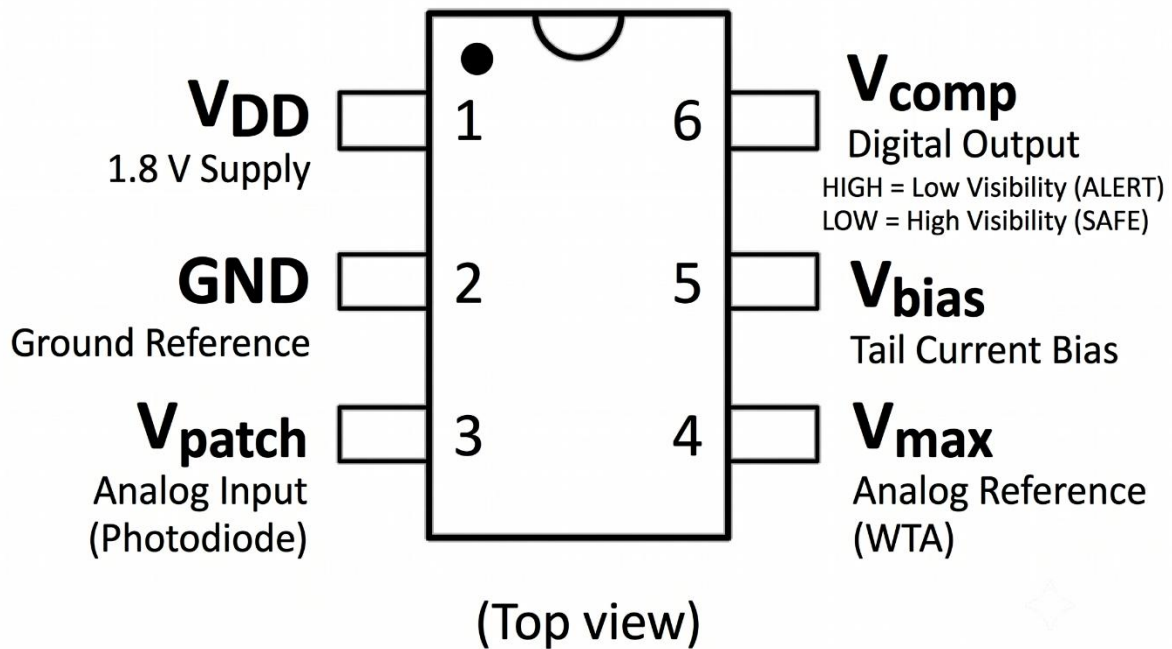
6. TARGET TECHNICAL SPECIFICATION

Parameter	Symbol	Target Value	Unit	Condition
Supply voltage	VDD	1.8	V	Nominal
Total power consumption	P _{total}	< 50	μW	All blocks active
Comparator static current	I _{SS}	~10	μA	M5 tail current
Threshold voltage	V _{thresh}	V _{max} / 2	V	Adaptive
Input voltage range	V _{in}	0 – VDD	V	Rail-to-rail
Propagation delay	t _{pd}	< 1	μs	10 mV overdrive
Output voltage HIGH	V _{OH}	> 1.6	V	LOW = high visibility
Output voltage LOW	V _{OL}	< 0.2	V	HIGH = low visibility
Input-referred offset	V _{os}	< 5	mV	With layout matching
Input-referred noise	V _n	< 10	mV RMS	1 Hz – 10 kHz
Technology node	—	0.5 μm / 0.15 μm	CMOS	—
Transistor count	—	9	MOSFETs	M1–M9
Operating temperature	T	0 – 70	°C	Commercial range
Battery life (CR2032)	—	> 267	days	39 μW continuous

7. Pinout Specification

Pin	Name	Direction	Description
1	VDD	Power	1.8 V supply
2	GND	Power	Ground reference
3	V _{patch}	Input (analog)	Photodiode signal (0 – VDD)
4	V _{max}	Input (analog)	Peak light voltage from WTA block
5	V _{bias}	Input (analog)	Tail current bias for M5
6	V _{comp}	Output (digital)	Comparator output (HIGH = low visibility)

8. Pin Configuration



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